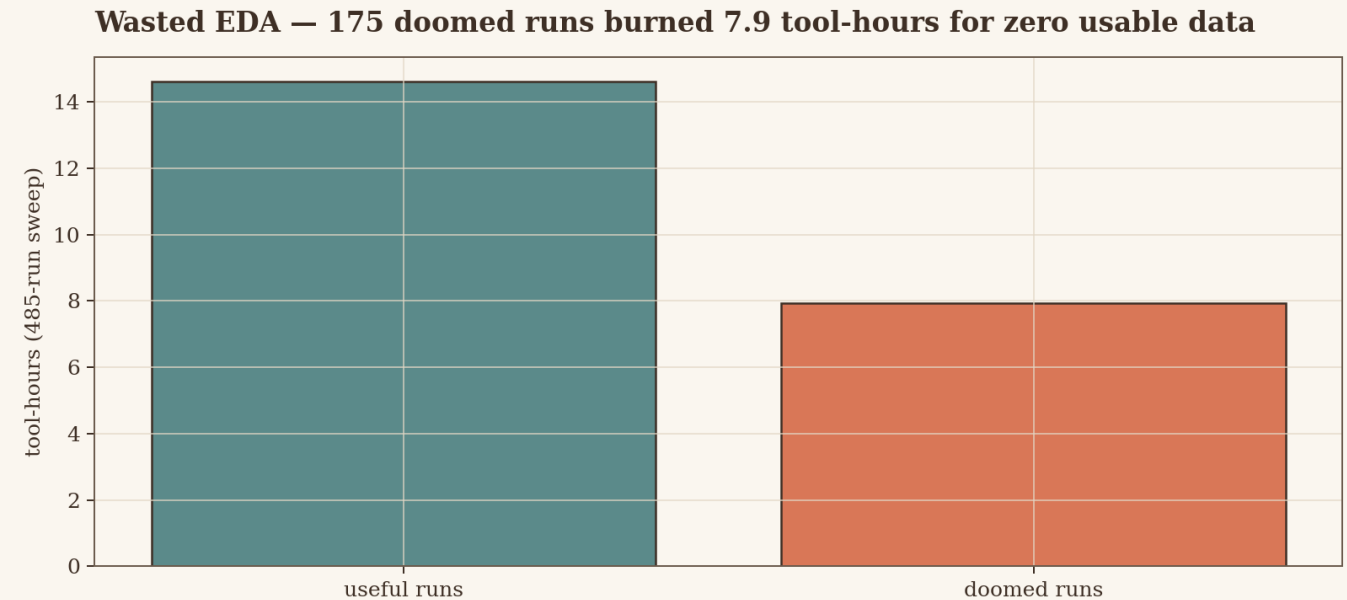


FlowGuard: stop burning silicon before it exists

- Feasibility-aware optimization for the open-source SKY130 flow
- Frozen 15.8 ns boundary + 5-way shootout · winner QoR 0.103
- Winner layout render committed; raw GDS remains server-local — Purdue CHIPSandAI

The wasted-EDA problem

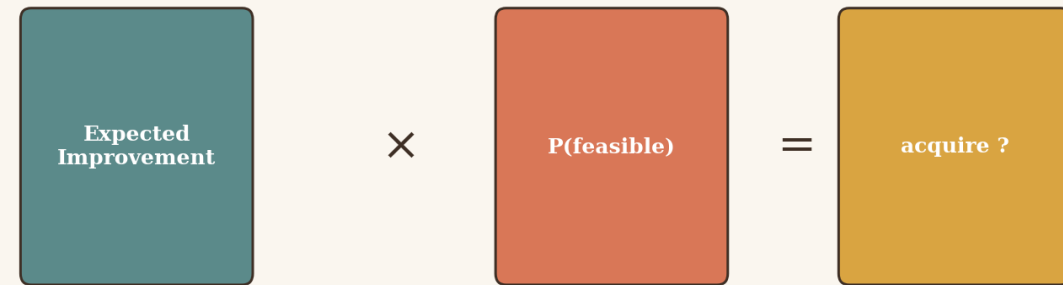
- 175 doomed runs burned 7.93 tool-hours for zero data
- Failures arrive late, at signoff — after the compute is spent
- Blind search re-samples dead regions instead of learning them



The FlowGuard idea: $EI \times P(\text{feasible})$

- Quality predictor $\times$ feasibility classifier — one score per candidate
- Calibration only with ≥ 3 obs/class, else declared uncalibrated
- Every pick logged: scores, seeds, hashes, model versions

FlowGuard idea — score every candidate as $EI \times P(\text{feasible})$

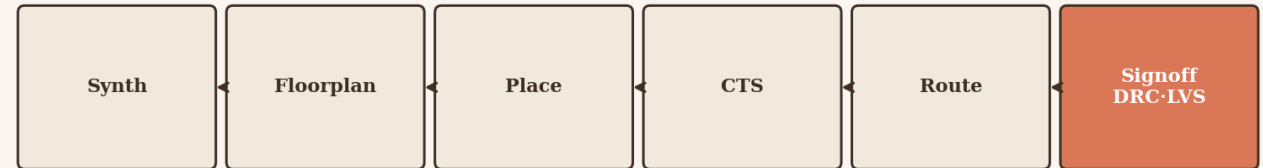


skip hopeless corners without spending a run — explore risky corners deliberately

A real SKY130 flow, not a toy

- LibreLane 3.0.14 · sky130A · 728-cell sensor-MAC stress block
- Feasible = SUCCESS + DRC 0 + setup/hold $\geq$ 0 + route 100% + LVS + signoff
- 72-candidate pool · 8 shared seeds + 16 adaptive picks per method

A real SKY130 flow — failures arrive late, after the compute is spent



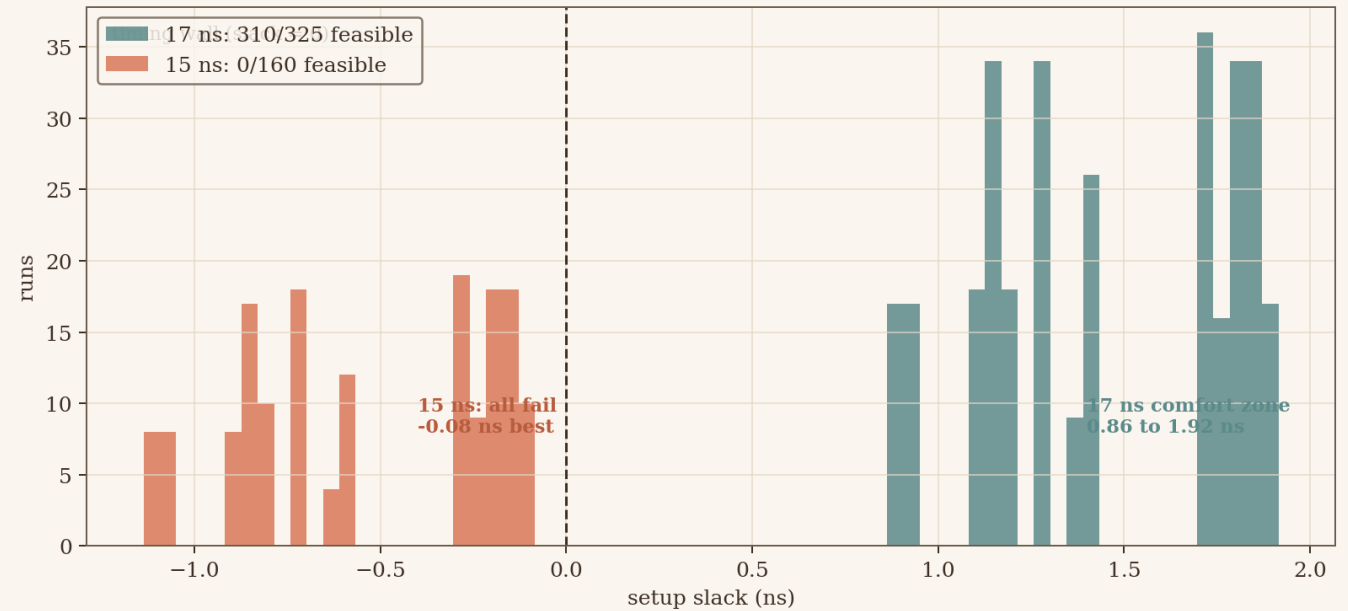
LibreLane 3.0.14 · sky130A · 728-cell sensor MAC · ~3 min per run

feasible = SUCCESS + DRC 0 + setup/hold $\geq$ 0 + route 100% + LVS + signoff

The 17/15 cliff

- 17 ns: 310/325 pass, slack 0.863–1.915 ns
- 15 ns: 0/160 pass — 159 × TIMING_FAIL
- No slope between regimes: hunt the boundary, don't guess it

The 17/15 cliff — setup slack distribution (485 sweep runs)

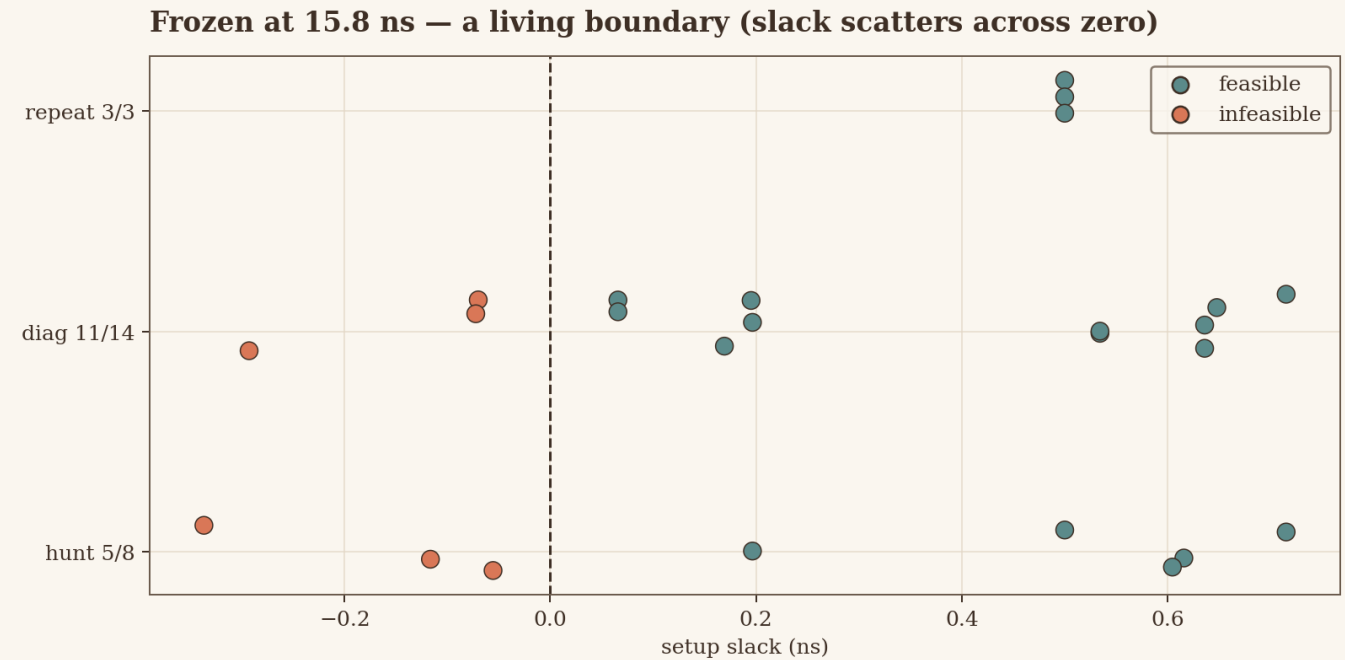


Frozen at 15.8 ns: a living boundary

■ Hunt 5/8 · diag 11/14 · repeat 3/3 identical

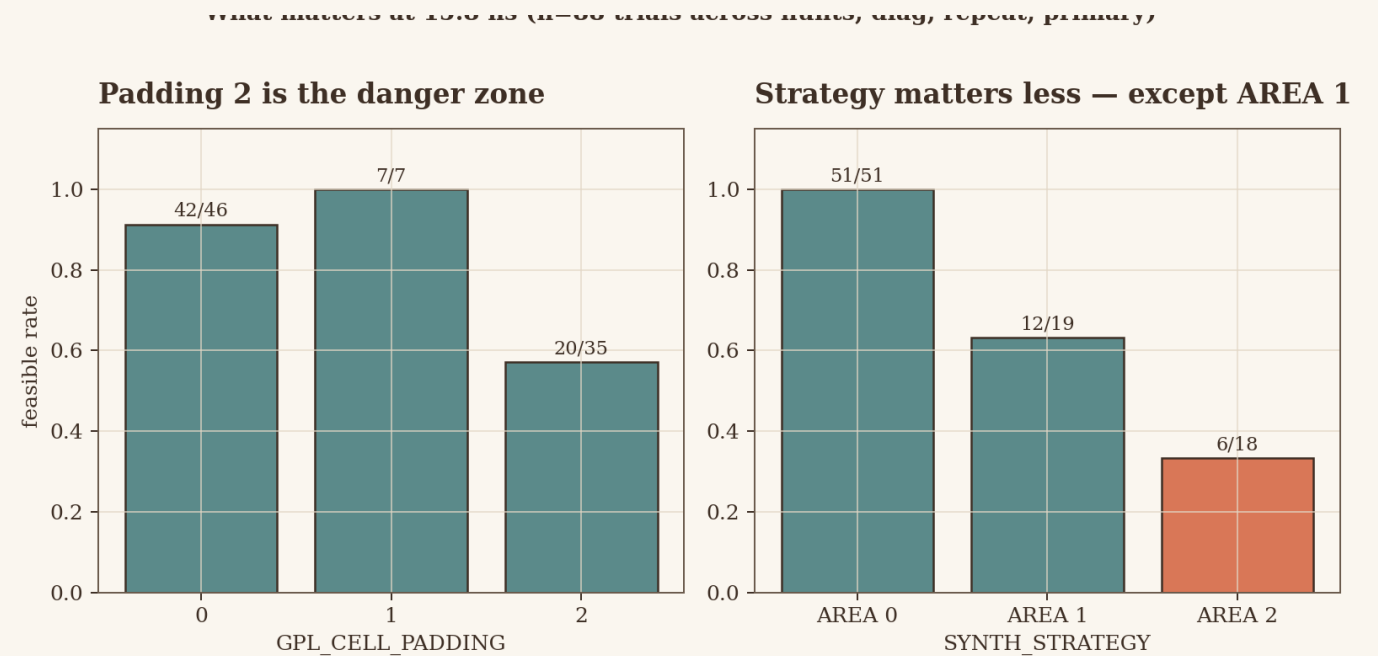
■ Slack scatters across zero (−0.34 to +0.72 ns): knobs decide

■ 3/3 identical repeats — signal, not tool noise



What matters: padding & strategy

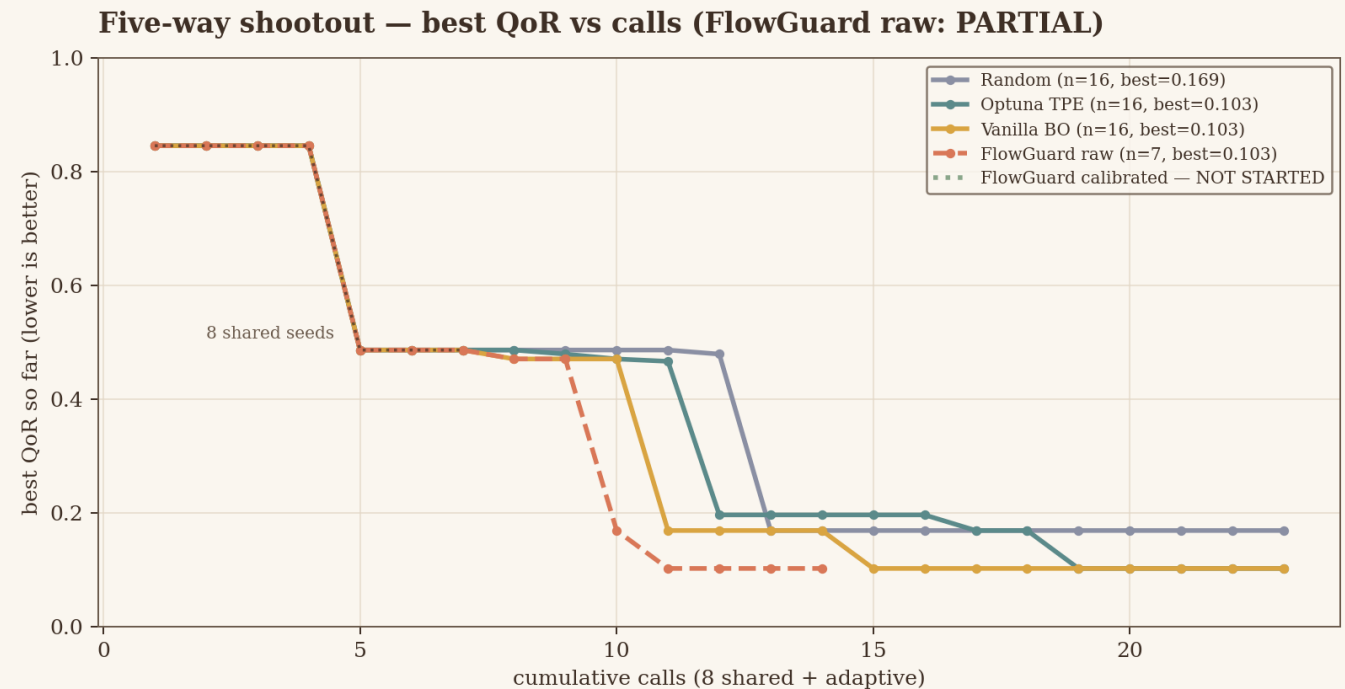
- Padding 2 fails far more often than 0/1 at 15.8 ns
- AREA 1 + high padding = the diagnostic grid's only failures
- Low padding + AREA 0 passes consistently (n=72 @15.8 ns)



A feasibility model should beat blind search here

Five-way shootout: best QoR vs calls

- Vanilla BO & Optuna → 0.103 · FlowGuard raw → 0.103 at adaptive call 11
- Random trails at 0.169 · FlowGuard raw recorded (11/16 feasible)
- FlowGuard calibrated: NOT STARTED (gated on calibration threshold)

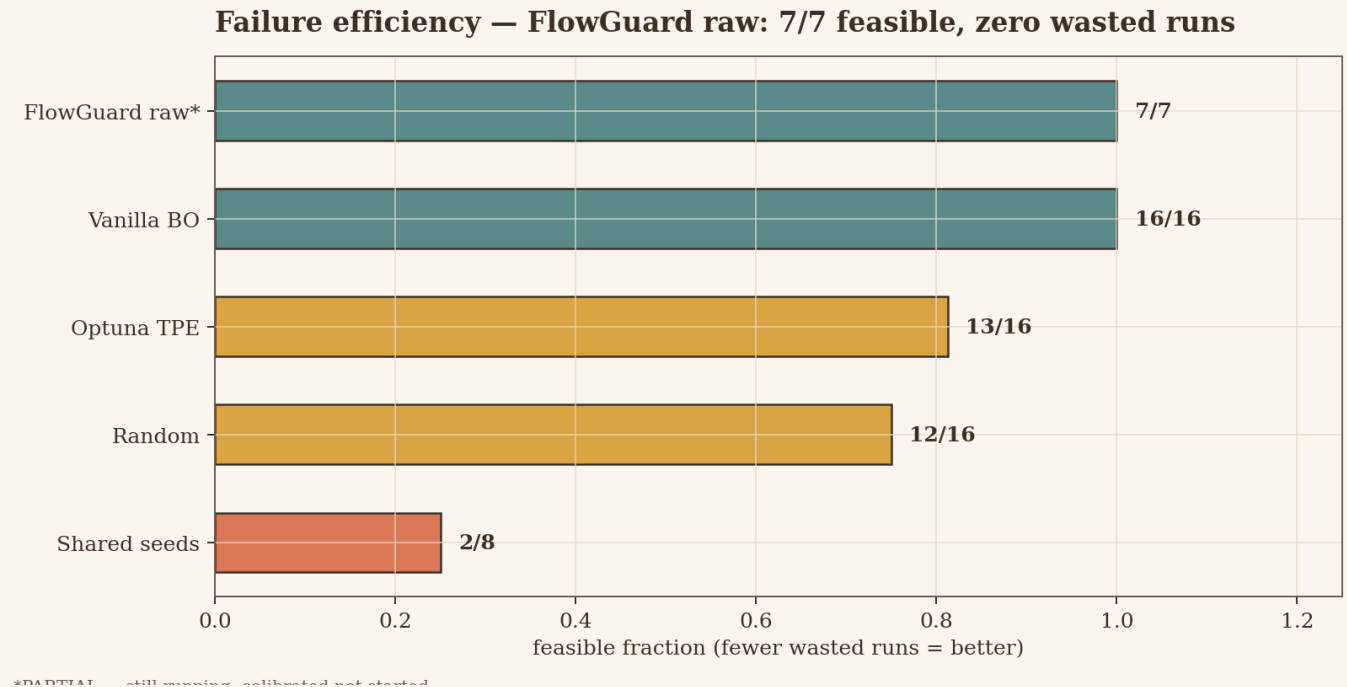


Failure efficiency

■ Vanilla BO 16/16 · FlowGuard raw 11/16 · Optuna 13/16
· random 12/16

■ Shared seeds only 2/8 — this does not support a
fewer-failures claim

■ Every avoided failure returns ~3 min of budget to
useful search



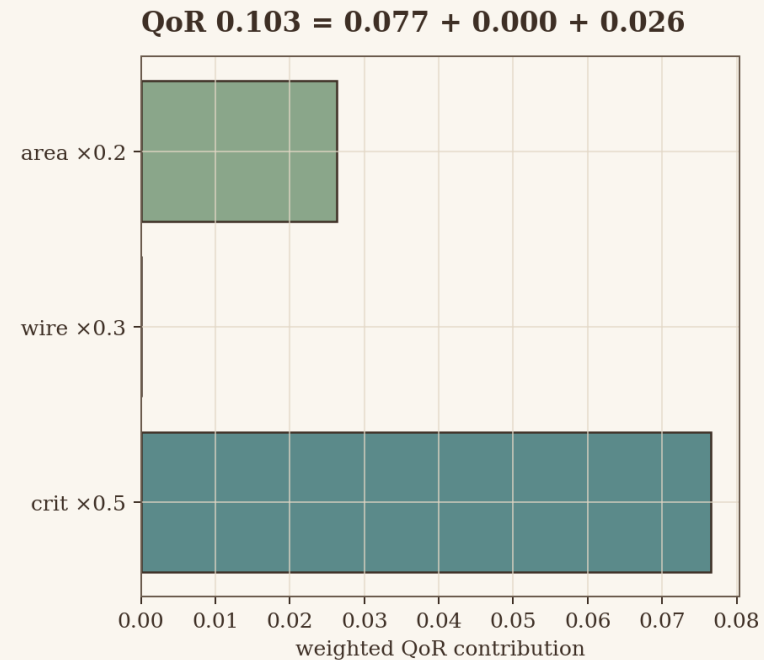
**PARTIAL — still running; calibrated not started*

The winner + GDS hero

■ Selected cand_014: QoR 0.103 · ws 0.6159 ns · hold 0.1122 ns

■ 15137 μm^2 · 27229 μm wire · pad 0 / dens 52 / grt 0.2 / AREA 0

■ No klayout here → metrics panel from signoff data:
pitch/renderers/FlowGuard-recovery__results__primary-optui



Winner: optuna cand_014 (call 19)

setup slack 0.616 ns

hold slack 0.112 ns

cell area 15137 μm^2

wirelength 27229 μm

DRC 0 · route 100% · LVS pass

pad 0 · dens 52 · grt 0.2 · AREA 0

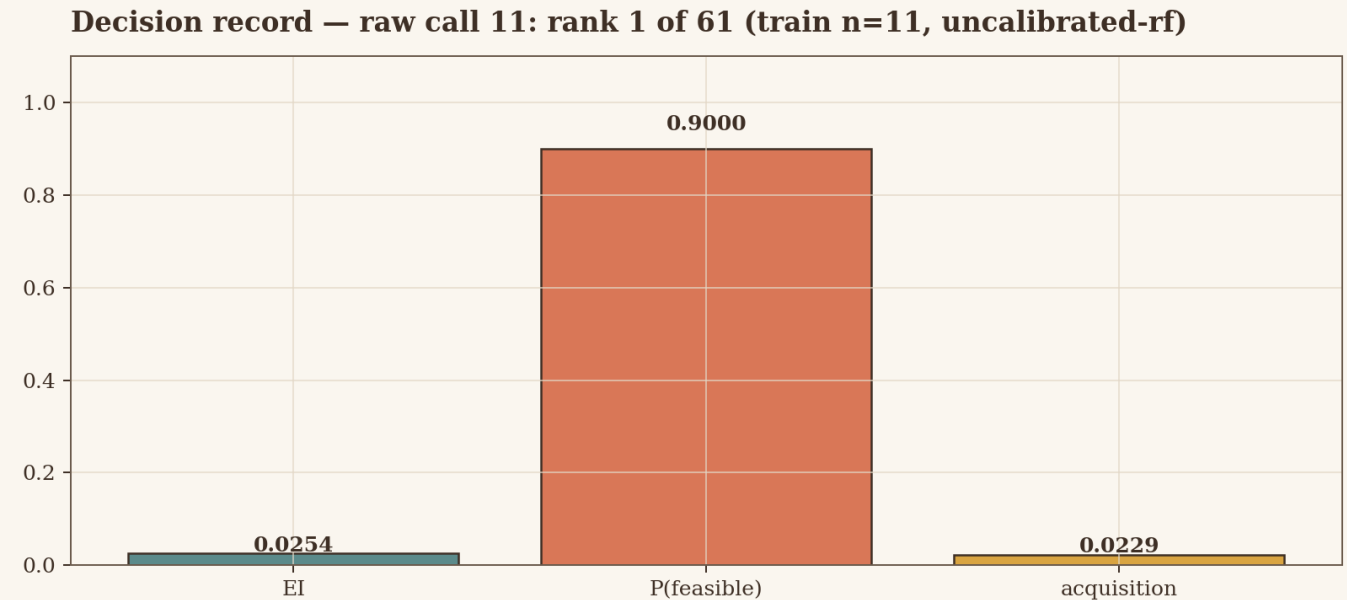
tress.gd

One AI decision, fully replayable

■ Raw call 11: rank 1 of 61 · EI 0.0254 · P(feas) 0.90

■ Seed 1340 · data 8dfeaf7d3fee... · uncalibrated-rf

■ Same hash + versions = same pick. Lore-free optimization.



Integrity + the ask

- Running traces marked recorded · calibrated not started · tile assumed 2×1, unvalidated
- Nothing extrapolated or cherry-picked — rerun builder refreshes all numbers
- Ask: collaborators + compute — finish calibrated, add a 2nd design, harden provenance